

Course Code: CN301

Course Name: Computer Networks

Semester: Fall 2026

Course Objectives:

Understand network architectures, protocols, routing, congestion control, and modern communication systems.

UNIT 1: Introduction to Computer Networks

- Network Models
- OSI Model
- TCP/IP Model
- Network Devices

UNIT 2: Data Link Layer

- Framing
- Error Detection
- Flow Control
- MAC Protocols

UNIT 3: Network Layer

- IP Addressing
- Subnetting
- Routing Algorithms
- IPv4 and IPv6

UNIT 4: Transport Layer

- UDP
- TCP
- Congestion Control
- Flow Control

UNIT 5: Application Layer

- DNS
- HTTP
- FTP
- Email Protocols

Suggested Books:

- Andrew Tanenbaum – Computer Networks
- Kurose & Ross – Computer Networking